

Research on Enhanced Small Object Detection in Autonomous Driving Based on an Optimized YOLOv11 Model

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Abstract

In autonomous driving scenarios, the difficulty in detecting small objects leads to low detection accuracy. This study proposes an improved method for vehicle and pedestrian detection based on an optimized YOLOv11 model. First, the DySample module upsamples and directly generates the coordinates of sampling points without using convolution kernels, reducing the computational complexity and guaranteeing ultra-lightweight and efficient performance. It proposes a new detection head, named Dynamic Alignment Detection Head (DADH) with task decomposition and feature alignment strategies, for more effective and precise object detection that raises the precision of small object detection and reduces the parameter count in the detection head. Lastly, it decreases network loss by using Inner-SIoU and accelerates regression convergence. Experimental results on the KITTI dataset demonstrate that the enhanced YOLOv11 model obtained 94.7% mAP_{0.5} and 94.1% mAP_{0.5-0.95}, with 2.1% and 1.9% gains over the baseline model, while the number of parameters is reduced by 15.4%. The detection precision of the proposed YOLOv11 model is improved and lightweight, which makes it suitable to run in resource-limited environments and highlights its practicality.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision problems; Object detection.

Keywords

Small Object Detection, DySample, DADH, YOLOv11, Inner-SIoU

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1 Introduction

With rapid development in the field of autonomous driving, the development of an accurate and real-time object detection system is very crucial for the safety of an autonomous vehicle on the road^[1]. Algorithms generally work fine for large vehicles with clear features. However, smaller objects-like pedestrians-present a problem. Smaller objects are more likely to lose features or get confused in real-world traffic. Autonomous vehicles also face challenges in dynamic object detection in resource-limited environments. In fact, the detection scenario is continuously modified by external factors, like speed and variable angles of view with respect to pedestrians or other vehicles. Thus, detection has become complicated in nature^[2]. Present techniques cannot adapt to changing conditions well enough. More omissions and false identifications might happen, further decreasing the accuracy with reduced safety by increasing the chance of an accident. Hence, occluded and small object detection has been one of the major fields of research, playing a key role in both technological advancements and real-world applications of autonomous systems^[3].

Motivated by such problems, Liang et al. developed the Edge-YOLO optimized for the small object detection in an autonomous driving scenario^[4]. It combines a pruned feature extraction network with a compressed fusion network to improve the accuracy of prediction at different scales. Liu Hui et al. also proposed the DPE YOLO^[5], which uses the K-means+D clustering algorithm to improve the detection accuracy for various objects^[6]. To solve the problem of missed detections for small objects, they integrated the PAA module^[7] that captures key details of these objects. However, both have major drawbacks: longer training times and slower detection speeds.

Despite some progress in detecting small and occluded objects in autonomous driving, challenges remain. These include low pixel resolution and blurred images, which impede feature extraction.

2 Proposed method

Based on the design of detection frameworks, Algorithms for detecting objects are commonly separated into two types: two-stage detection and single-stage detection^[8]. Two-stage methods generally offer higher accuracy but at the cost of slower detection speeds. The second type, single-stage methods, uses regression-based detection to directly output object classifications and bounding box regressions in a single forward pass^[9]. Prominent algorithms in this category include the YOLO series, SSD, and Retina Net. Single-stage methods excel at performing object detection tasks quickly, making them ideal for real-time scenarios like live image capture and

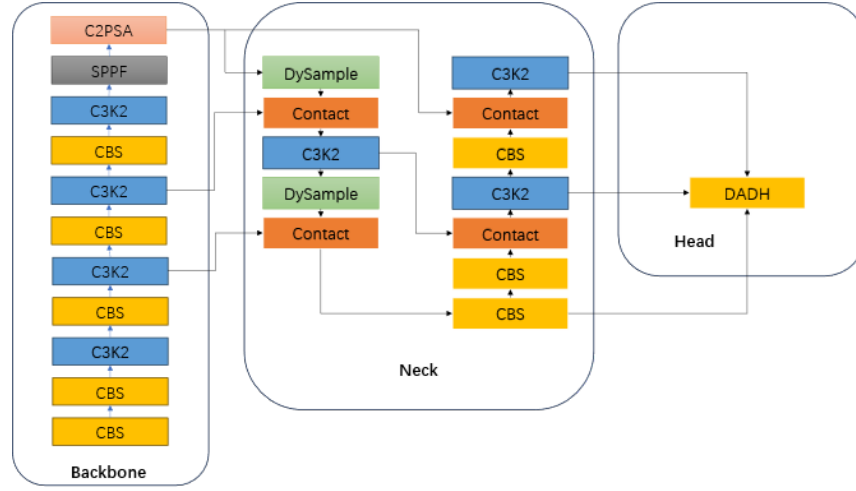


Figure 1: The improved YOLOv11n model.

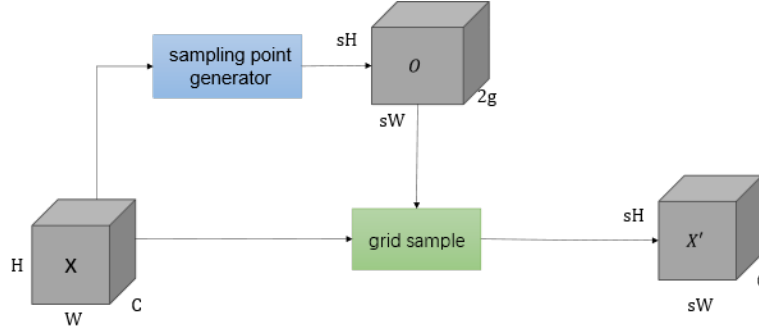


Figure 2: Sampling-based dynamic upsampling.

autonomous driving. The architecture of the improved YOLOv11n model is shown in Figure 1.

2.1 DySample

The upsampling method adopted in YOLOv11 is NN interpolation, which, though computational, is fixed in its behavior and hence cannot adapt to any variation in the content. This rigidity causes much limitation of flexibility and reduced adaptability for complicated tasks. The concept of DySample was introduced to overcome such challenges.

The basic idea of DySample is point sampling. DySample improves upsampling in controlling initial sampling positions, refining the range of offset movements, and segmenting upsampling into multiple independent groups^[10]. In Figure 2, the grid sampling process of DySample is shown.

Then the feature map X is resampled by the shown sampling points S with PyTorch's `grid_sample` function. The resampled feature map is hence the upsampled feature map. Such a paradigm shift from convolution to sampling simplifies not only the process but also avoids redundant computation associated with dynamic convolution kernels, realizing notable efficiency gains.

Earlier designs used the same grid points to initialize all sampling points, and this always caused their models to be sparsely distributed and poorly performing. Figure 3 DySample uses bilinear initialization for an equal distribution of points around the center of the grid, and a static range factor, such as 0.25, is used to keep the distribution of the sampling set much localized:

$$O = 0.25 \times \text{linear}(X) \quad (1)$$

For more flexibility, DySample further introduces a dynamic range factor that enables the offset range to change dynamically according to the content. Another linear layer outputs a dynamic range with a range of $[0, 0.5]$. σ is the sigmoid function. The principle of designing the dynamic sampling set generator can be seen from Figure 4.

$$O = 0.5 \times \sigma(\text{linear}_1(X)) \times \text{linear}_2(X) \quad (2)$$

DySample divides the feature map channels into multiple groups, where each group shares a set of offsets. The proposed grouping strategy balances the reduction of computational cost with the preservation of the expressiveness of the sampling process, which yields substantial performance gains across diverse tasks.

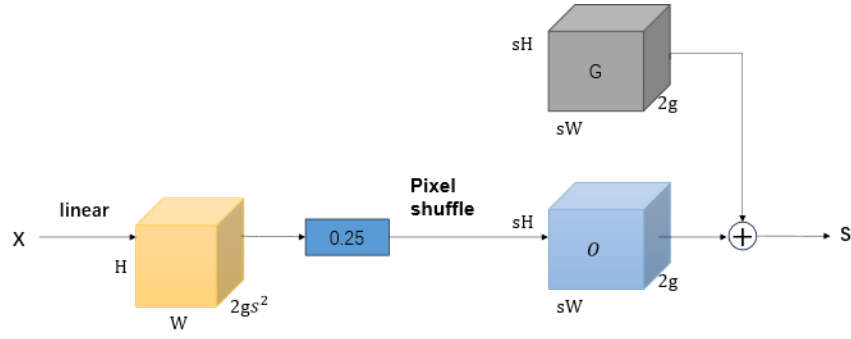


Figure 3: Static sampling set generator.

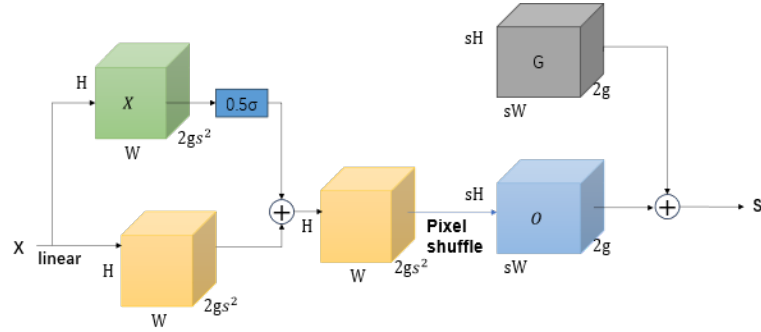


Figure 4: Dynamic sampling set generator.

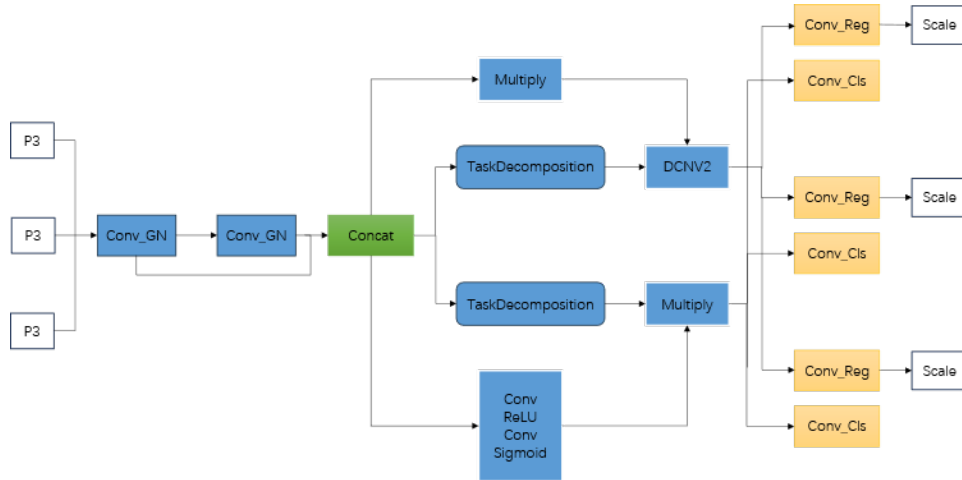


Figure 5: Architecture of Dynamic Alignment Detection Head.

2.2 DADH

In YOLOv11, the underlying decoupled object detector mainly relies on the classification and localization branches being separated, interaction between which is inherently inhibited. While one classification branch predicts the categories of objects, the other focuses on the object position. However, without appropriate communication and collaboration of both branches, the ultimate localization

accuracy, especially for the cases involving complexity, is usually not optimized and often leads to several false positives and missed detections.

This paper presents a novel Dynamic Alignment Detection Head, named DADH, to enhance the conventional YOLOv11 detection head, as illustrated in Figure 5. It presents task decomposition and feature alignment methods to offer efficient object detection with

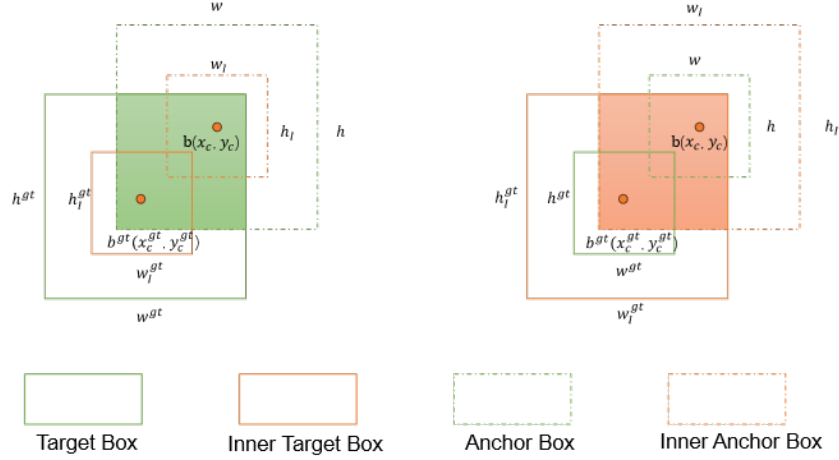


Figure 6: The structure of Inner-SIoU.

improved accuracy. It mainly consists of several key modules: a shared convolutional layer, task decomposition module, dynamic convolution alignment module, and class probability generation module. These modules interact with each other in order to provide efficient feature extraction and object detection.

The first layer is the shared convolutional layer, which extracts preliminary features from the input feature map. Since GN has been proved in the FCOS paper to improve the performance of detection heads for both localization and classification, this work introduces the GN-Conv module, which replaces traditional BN with GN in normal convolution. Though BN is widely used for normalization in deep learning and plays an important role in improving the training and convergence speed, it still has some limits. GN overcomes BN's dependence on batch size and can achieve better performances and generalizations than BN in many tasks.

To improve the registration accuracy of the regression features, introduce a dynamic convolution alignment module based on the two regression features. This module implements a convolution layer to generate spatial offset and mask matrices that adapt the regression features by dynamic convolution alignment. These generated offsets and masks are applied for changing the location and nonrigid shape of the convolution kernel so that features can be precisely aligned. For category feature alignment, two convolutional layers generate the class probabilities. The first convolutional layer reduces the count of channels of the feature map to a quarter and applies ReLU activation. Another convolutional layer generates the final class probabilities that are normalized using the sigmoid function. Then, the corresponding convolutional layers treat the regression and category features separately for the bounding box regression values and category predictions, respectively. These operations, summarily named Dynamic Feature Selection with interactive features, would enhance object detection precision.

In particular, the DADH enhances object detection accuracy through task decomposition and dynamic convolution alignment. Meanwhile, it maintains computational efficiency and reduces the parameter count in the detection head.

2.3 Inner-SIoU

In this paper, Inner-SIoU instead of CIoU is used as the loss function in YOLOv11 to improve the convergence speed. The structure of Inner-SIoU is presented in Figure 6.

The GT box and anchor box are represented by b^{gt} and b , respectively. The centroids of the GT box and its inner region are represented by (x_c^{gt}, y_c^{gt}) and (x_c, y_c) , respectively. Height of GT box and anchor box are h^{gt} and h , and width of GT box and anchor box are w^{gt} and w . Usually, the variable ratio for scaling factor is within $[0.5, 1.5]$. Derivation formula as follows:

$$b_l^{gt} = x_c^{gt} - \frac{w^{gt} * \text{ratio}}{2}, b_r^{gt} = x_c^{gt} + \frac{w^{gt} * \text{ratio}}{2} \quad (3)$$

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} * \text{ratio}}{2}, b_b^{gt} = y_c^{gt} + \frac{h^{gt} * \text{ratio}}{2} \quad (4)$$

$$b_l = x_c - \frac{w * \text{ratio}}{2}, b_r = x_c^{gt} + \frac{w * \text{ratio}}{2} \quad (5)$$

$$b_t = y_c - \frac{h * \text{ratio}}{2}, b_b = y_c^{gt} + \frac{h * \text{ratio}}{2} \quad (6)$$

$$\text{inter} = \left(\min(b_r^{gt}, b_r) - \max(b_l^{gt}, b_l) \right) * \left(\min(b_b^{gt}, b_b) - \max(b_t^{gt}, b_t) \right) \quad (7)$$

$$\text{union} = h^{gt} * \text{ratio}^2 + w^{gt} * w * h * \text{ratio}^2 - \text{inter} \quad (8)$$

$$IoU^{\text{inner}} = \frac{\text{inter}}{\text{union}} \quad (9)$$

$$L_{\text{Inner-SIoU}} = L_{\text{SIoU}} + IoU - IoU^{\text{inner}} \quad (10)$$

Combining SIoU with Inner-IoU to form Inner-SIoU has significantly improved the convergence speed of the model. Combining them improves the model's performance in handling IoU samples that are imbalanced and improves generalization and performance.

Table 1: Results of ablation experiments on KITTI dataset.

Model	DySample	DADH	inner-SIoU	mAP@0.5	mAP@0.5:0.95	Parameter/10 ⁶	GFLOPS
1				0.926	0.722	2.6	6.3
2	✓			0.939	0.739	2.6	6.3
3		✓		0.931	0.721	2.2	6.9
4			✓	0.933	0.731	2.6	6.3
5	✓	✓		0.941	0.732	2.2	6.9
6	✓	✓	✓	0.947	0.741	2.2	6.9

3 Experiments

3.1 Dataset

Performance for the proposed model was measured, which was trained and evaluated using the KITTI dataset. During preprocessing, the classes that were kept included Truck, Van, Tram, and Cyclist; Pedestrian was changed to Person, and all others were excluded. This dataset consists of 7,481 images, which have been divided into training, validation, and test sets in a ratio of 7:1:2.

3.2 Experimental setup

Experiments were performed on a system with Windows 11, one NVIDIA GeForce RTX 4060 GPU with 12 GB of memory, and deep learning frameworks PyTorch 2.0.1 and CUDA 11.8. The baseline model is YOLOv11n, whose input image size is 640×640. The SGD optimizer is adopted in training, where the momentum is 0.9, the initial learning rate is 0.01, the weight decay coefficient is 0.005, the batch size is 16, and the number of epochs is 300 in total.

3.3 Evaluation Metrics

Precision, Recall, Mean Average Precision, the number of parameters, and GFLOPS are some of the metrics used in the performance evaluation of the different detection models. Further, the mathematical definitions of these metrics are underlined:

$$P = \frac{T_P}{F_P + T_P} \quad (11)$$

$$R = \frac{T_P}{F_N + T_P} \quad (12)$$

$$AP = \int_0^1 P * RdR \quad (13)$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (14)$$

Here, T_P is the number of true positive samples that are correctly classified, T_N represents the amount of true negative samples that are correctly identified; F_P represents those false positive samples that were mislabeled as negative; whereas F_N represents those false negative samples that were incorrectly labeled as positive. AP for a class is done by taking the area under the Precision-Recall graph, with Recall on the x-axis and Precision on the y-axis.

3.4 Ablation Study

Ablation experiments on the KITTI dataset were conducted to evaluate each module's enhancements.

According to the data in Table 1, it is evident that all improvements are beneficial for better model detection. By using a baseline YOLOv11n, changing nn upsampling module for DySample provided 1.3% and 1.7% improvement in mAP@0.5 and mAP@0.5:0.95 respectively, with almost unnoticeable differences in a parameter count or GFLOPS. In general, using only the new detection head DADH increased mAP@0.5 by 0.5%, while the mAP@0.5:0.95 has decreased slightly by 0.1%. Besides, the parameter count was reduced by 0.4, while GFLOPS increased by 0.6. Using inner-SIoU instead of Ciou independently allows improvements of 0.7% in mAP@0.5 and 0.9% in mAP@0.5:0.95 with negligible changes in GFLOPS and parameter count.

Integrating DySample with DADH improved mAP@0.5 by 1.5% and increased mAP@0.5:0.95 by 1%. The parameter count reduced by 0.4, while GFLOPS increased by 0.6. Furthermore, inner-SIoU further improved mAP@0.5 performance by 2.1%, with mAP@0.5:0.95 improving by 1.9%, in tandem with a consistent reduction of 0.4 in the parameter count and increment of 0.6 in GFLOPS. From these results, one can see that every module improves mAP. The final model is not only overwhelmingly better in accuracy but has also reduced 15.4% of parameters compared to the baseline model, hence arriving at a lightweight design. In general, the superiority varied in degree over the baselines.

3.5 Comparative Experiments

To further verify the proposed enhancements, using the same dataset and experimental configuration, the model was benchmarked against six established network architectures: SSD, YOLOv5s, YOLOv8n, YOLOv10n, YOLOv11n, and RT-DETR-r18. The results are shown in Table 2. Compared with SSD, YOLOv5s, YOLOv8n, YOLOv10n, YOLOv11n, and RT-DETR-r18, the improved YOLOv11n raised mAP@0.5 by 18.5%, 8.1%, 4.1%, 3.2%, 2.1%, and 2.8%, and improved mAP@0.5:0.95 by 22.5%, 12.7%, 6.2%, 4.3%, 1.9%, and 6.8%, respectively. For small object detection, the improved YOLOv11n outperformed all other algorithms on both mAP@0.5 and mAP@0.5:0.95. Results have shown improvements in the performance of the enhanced YOLOv11n model, which could allow autonomous vehicles to better detect pedestrians and other vehicles, even in constrained resource environments.

4 Conclusion

This work proposes an improved YOLOv11 model to overcome the challenges due to small object detection. First, DySample replaces the nearest-neighbor interpolation method. Then, a new Dynamic Aligned Detection Head (DADH) is proposed. At last, traditional

Table 2: Performance comparison of detection algorithms on.

Model	mAP@0.5	mAP@0.5:0.95	Parameter/ 10^6
SSD	0.762	0.516	26.5
YOLOv5s	0.866	0.614	7.02
YOLOv8n	0.906	0.679	3
YOLOv10n	0.915	0.698	2.76
YOLOv11n	0.926	0.722	2.6
RT-DETR-r18	0.919	0.673	19.9
Ours	0.947	0.741	2.2

CIoU is replaced by Inner-SIoU. Comparative analysis with the baseline model and other existing approaches has shown that the proposed YOLOv11n model improves small and occluded object detection significantly. In comparison with the base model and other existing approaches, the proposed algorithm has significantly improved in the detection of small and occluded objects. This could involve further computational cost reduction with pruning, knowledge distillation, or even improving detection speed and adapting better to the demands of an autonomous driving model in the future.

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